

An Automatic Path Planning Method of Pedicle Screw Placement Based on Preoperative CT Images

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Abstract—Due to good fixation and obvious clinical advantages in the treatment of spinal diseases, pedicle screw technology is widely employed in spinal surgery. However, most screw placement paths are manually selected, which is a complicated and time-consuming process. Therefore, various kinds of spine surgery robots have been invented to assist during surgeries. An automatic path planning method of pedicle screw placement is proposed in this article. First, based on preoperative CT scans, the spine of the patient can be segmented and each individual vertebra is classified by using a trained deep learning network model. Additionally, a local coordinate system in every single vertebra is established according to its anatomical characteristics. Second, after analyzing the images to recognize the feature points, the pedicle region and route of screw placement are identified, which renders completion of the surgical path planning automatic. The experimental results show that 93% of the planning results can be directly applied in surgeries after surgeon evaluation. In addition to the improvement in accuracy, the whole algorithm does not rely on any human assistance. This fully automatic process greatly reduces the time required for surgical planning.

Index Terms—3D image processing, spine surgery robot, pedicle screw placement, automatic path planning, deep learning.

I. INTRODUCTION

THE INCIDENCE of spinal diseases has gradually increased in recent years, becoming one of the major diseases that affect people's health. Among these spinal diseases, intervertebral disc herniation, spinal degeneration, spinal spondylolisthesis and vertebral fracture are the most common diseases in clinical observation and usually require screw fixation to complete surgical treatment [1]. Manual pedicle screw placement is generally employed to complete the fixation of the spine in the process of traditional surgical treatment. Screw slippage may cause serious clinical complications, such as neurovascular injury, dural sac injury, and organ and muscle injury. In addition, the incidence of complications of various diseases due to improper operations and other factors can reach 52.58% in traditional spine surgeries [2], [3], [4]. Therefore, in the past decade, certain spinal surgery robots focused on assisting pedicle fixation have been designed to replace the manual screw placement of a surgeon [5], and certain studies on medical screws have been performed [6]. These advancements help to complete lumbar repair and fixation more conveniently and accurately, which reduces not only the operational burden of surgeons but also postoperative complications as much as possible and improves the success rate of surgery. The key point of pedicle screw fixation is to accurately determine the position and angle of the screw path for drilling in the vertebra. However, due to the limited field of vision in surgeries, surgeons need to use CT imaging equipment to determine the relative positions of the surgical instruments in the vertebra to ensure accuracy [7], [8]. Frequent CT scan operations will increase the exposure time of surgeons and patients to radiation and cause damage to their bodies. Therefore, the spinal surgery robot-assisted system can effectively reduce the number of CT (Computed Tomography) scans needed by using preoperative data to complete the placement planning of pedicle screws and can assist surgeons in completing high-precision and high-stability surgical operations. For example, Smith *et al.* presented a new, commercial robot-based, image-guided surgery system that automates the process of bone preparation and polyaxial screw placement. The system is based on the KUKA LBR iiwa 7-axis collaborative robot and uses

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CT images to calculate screw locations, which achieves good performance in spinal surgeries [9].

To improve the safety of pedicle screw placement, many methods of automatic planning for pedicle screw paths have been proposed and applied to robot-assisted systems. The development of fluoroscopic guidance and navigation technology enables operations to be completed with a preoperative planning path [10], [11], [12], and the automatic planning method greatly simplifies complicated operation workflow and improves the accuracy of operations with the help of 2D-3D image registration for intraoperative guidance [13], [14], [15]. Therefore, preoperative planning has become the key to simplifying a complicated process and to improving the accuracy of pedicle screw operations. Helm *et al.* suggested that imaging- and computer-aided planning is becoming an important research direction of spine surgery because of its high accuracy [16], [17], [18]. Lee *et al.* utilized CT images that contain lumbar pedicles as input to manually finish segments of vertebrae. They recovered the vertebral geometry through two-dimensional dynamic threshold segmentation with local optimal parameters and then extracted the vertebral pedicle region edges according to the obtained segmental vertebral bodies and spinal canal [19], [20], [21]. Wicker *et al.* performed operations in two dimensions in the coordinate system of medical imaging data. For each image slice, the minimum pedicle width is identified, and examination of all slices identifies the overall minimum pedicle width. The optimum trajectory is determined using a 3D linear least squares fit to the array of minimum pedicle width midpoints [22].

However, in these methods, vertebral information needs to be manually selected from CT images by researchers, which is extremely inconvenient. Some methods for segmentation and recognition of vertebrae have been proposed to complete complicated tasks [23], [24], [25]. Attanasio *et al.* proposed a spatiotemporal neural network based on U-Net that is aimed at segmenting soft tissues in endoscopic images and at enhancing the segmentation accuracy [26]. Sekuboyina *et al.* employed a simple multilabel, 2D, fully convolutional neural network (FCN) for segmentation of only the lumbar vertebrae [27]. Chen *et al.* detected vertebrae with a convolutional neural network trained with a specialized loss term to distinguish neighboring vertebrae [28]. To reconstruct the 3D model of vertebrae, both segmentation and identification need to be achieved [29].

Furthermore, during the drilling operation, Jiang *et al.* planned the screw path by tracking and monitoring the force and position information of bone layer grinding in the simulated breathing state, achieving a success rate of 95% [30], [31], [32]. Knez *et al.* extracted the geometric properties of the vertebral structure through three-dimensional parameterized modeling of the vertebra and pedicle and combined them with structural properties to calculate the screw size and trajectory with CT image information. Further work has been performed to maximize the overlap between the screw and the bone area in the CT images to increase the strength of the screw to accommodate different spinal curvatures. Their method was eventually evaluated on 81 pedicles,

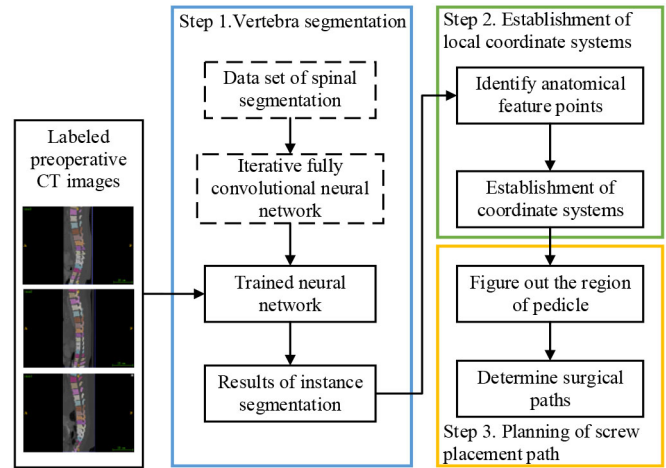


Fig. 1. Steps of automatic path planning of pedicle screw placement.

and screw tightness increased by $48 \pm 26\%$ in 99% of the pedicles, achieving remarkable results [33], [34].

These methods of automatic pedicle screw path planning often require manual operation to obtain a single vertebra in spine images. Most of these methods are applied according to the statistical regularity for path planning, which cannot achieve fully automatic processing and will cause problems when addressing the specificity of different vertebrae.

In this paper, a highly automatic pedicle screw path planning method is proposed based on preoperative CT images to solve the problems related to the complicated procedures and poor universality of the path planning methods in current pedicle surgeries. The main contributions of this paper include: (1) instance segmentation of CT images of the spine was completed by using a trained neural network model, enabling the clear demarcation of different instances of different vertebrae. (2) For each individual vertebral segment, a screw path of the pedicle is determined based on anatomical characteristics to obtain different routes suitable for vertebrae in different shapes, which improves the universality and accuracy of the planning method.

II. METHOD

Automatic path planning for the pedicle screw is based on the CT scans from which individual vertebra data are obtained, and points of screw placement paths are determined according to its anatomical features. The whole process consists of three steps, as shown in Fig. 1. First, instance segmentation of the vertebra was completed to ensure that each vertebra was recognized in the preoperative CT scan images of patients. In this step, based on a trained convolutional neural network model, the instance segmentation process is carried out so that the data of each single vertebra in the CT image are extracted. Second, a local coordinate system is established for each vertebra. Based on the image information of a single vertebra, an algorithm is designed to identify its anatomical geometric characteristic information. A local coordinate system can be established to help conveniently describe the vertebra. Third, route points of screws are determined, and planning paths are obtained. After rotation and translation of the vertebra, a surgical path can be calculated by identifying special points

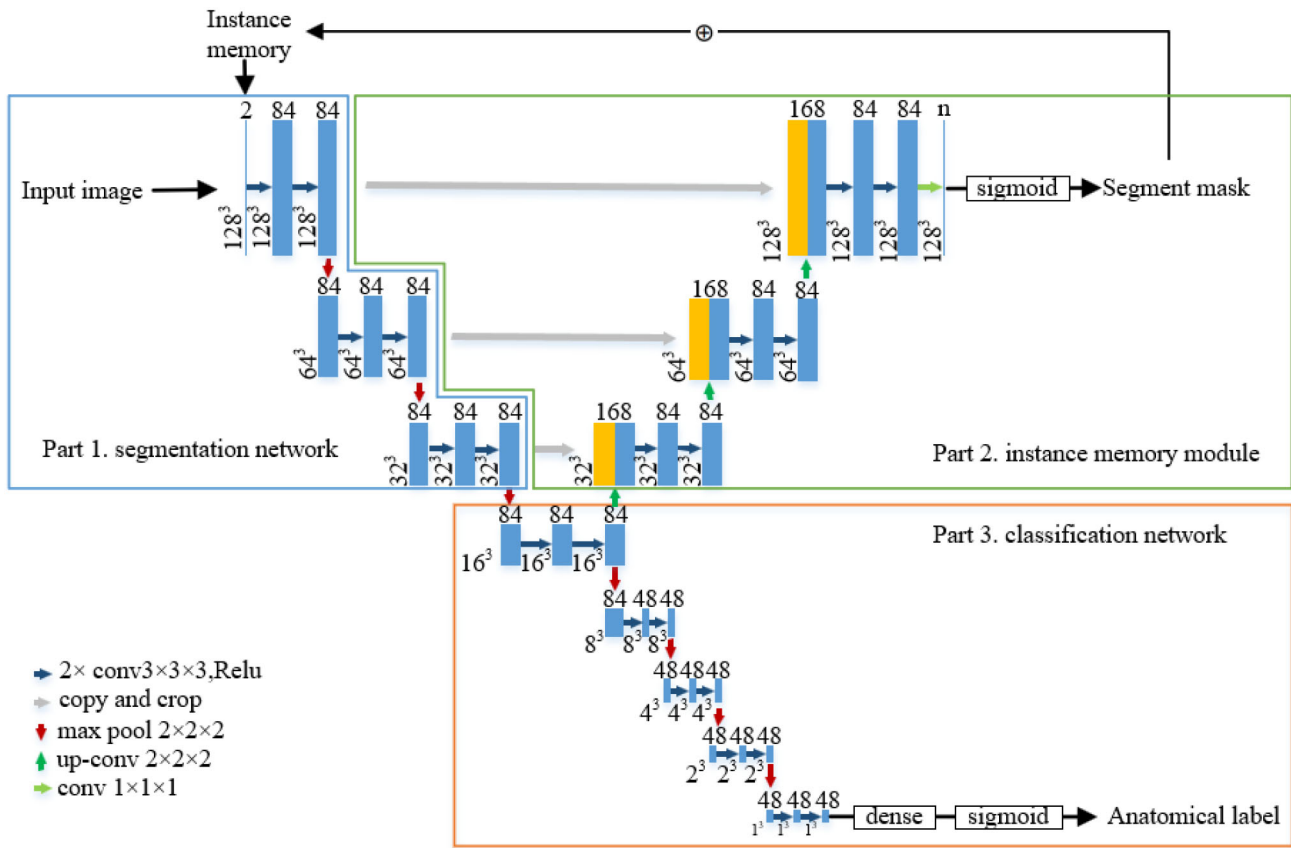


Fig. 2. Convolutional neural network architecture.

of the path related to pedicle screw placement. The steps of the automatic path planning method of the pedicle screw are shown in Fig. 1.

A. Vertebra Segmentation Based on a Neural Network

In spinal surgery, the segmentation and recognition of vertebrae is always a prerequisite for automatic analysis. The application of many computer-aided surgical methods cannot occur without recognition and extraction of single vertebral body data. Presently, the division of different segments of the vertebra is mainly selected and judged manually by surgeons. The complex operation process and immense error render the subsequent geometric feature recognition algorithm for the vertebra less effective. In recent years, an increasing number of neural network models have been applied to image segmentation in the medical field due to their good performance in classification problems. Among these methods, convolutional neural networks, as supervised machine learning models, have strong adaptability in different situations and can identify local features to classify them according to global training data. Therefore, this paper uses an instance segmentation method that is based on a convolutional neural network training model to complete the segmentation of a single vertebra in preoperative CT scan data.

1) *Convolutional Neural Network Model Design:* Inspired by the network architecture of 3D U-NET, a neural network training model is established as shown in Fig. 2. The whole

network model is mainly composed of three parts. The main part is a segmentation network, which segments vertebrae from the volume data of 3D images. The second part is an instance memory module that informs the network about already segmented vertebrae. This module is again input into the segmentation network with the input dataset, making the whole segmentation process start from top to bottom and helping the network to focus on the next vertebra that needs to be segmented in the next iteration process. The third part is a classification network that is added to distinguish fully visible vertebral areas. The number above each graph block in the figure represents the number of channels of the feature image, and the number on the left of the graph block represents the size of the feature image. For example, the data in the input layer are composed of an input data block and an instance memory data block to form 128 × 128 × 128 voxel blocks containing 2 channels, which are then transformed by the convolution operation into 128 × 128 × 128 voxel blocks containing 84 data channels. Afterward, the labeled sample image is input into the network model for training through this input layer.

In this network model, not only the original data with labels but also the instance segmentation data obtained in the last training are concurrently input. The instance memory patch is generated by prediction and then reiterated as input to optimize the next calculation. After the convolutional process, the input image is converted to a one-dimensional vector at the last layer through dense operation; this vector is utilized to complete

the classification and labeling process. The hidden layers of this model include convolution operations, normalization operations, activation operations and upwards sampling. In each layer, a $3 \times 3 \times 3$ block is selected to complete the two convolution operations, and then a pooling operation is added to ensure the invariance of the data, reducing the parameters and calculation while retaining the features and achieving dimensionality reduction to prevent overfitting. Simultaneously, in each layer, a nonlinear unit is added with the ReLu activation function, which not only reduces the overfitting and produces more efficient gradient descent and back propagation but also avoids the problem of gradient explosion and the disappearance of the gradient. Multilevel convolution and pooling operations improve the depth of the network and dig out the information in a deeper dimension of images. The weight and bias in the convolution process are iteratively adjusted by the whole network during backpropagation to achieve a smaller loss function and to optimize the classification effect. Another branch of upwards sampling is introduced into the network according to the U-Net framework while performing convolution operations. Through the upwards sampling process, an instance memory module with the same size of input dataset, including the information of labeled vertebrae already segmented, is obtained to predict the next vertebra.

On the other hand, a one-dimensional vector is obtained as the output layer of the neural network after continuous convolution, and a sigmoid activation function is applied to complete the classification and labeling of vertebrae.

2) *Loss Function and Network Training* : The network model is used to predict the labels of all the voxels in the input 3D volume data, so the calculation of the loss function is not only based on a single output value but also related to the volume data of the classified output labels. The calculation of the loss function includes the segmentation error and classification error; its equation is shown as follows,

$$\begin{cases} \mathcal{L} = e_{segment} + e_{classify} \\ e_{segment} = \lambda \cdot FP_{soft} + FN_{soft} \\ e_{classify} = -t \log p - (1 - t) \log(1 - p) \end{cases} \quad (1)$$

where, $\mathcal{L}$ is the loss function, which includes segment error $e_{segment}$ and classification error $e_{classify}$. The classification error $e_{classify}$ is mainly calculated from truth tags t (values 0 or 1) and probabilistic predictions p . The segmentation error $e_{segment}$ is mainly influenced by false positives FP_{soft} and false negatives FN_{soft} , which can be calculated using the following equations. λ represents the weight, and λ is chosen as 0.1 in the experiment.

$$\begin{cases} FP_{soft} = \sum_{i \in I} \omega_i \cdot (1 - t_i) p_i \\ FN_{soft} = \sum_{i \in I} \omega_i \cdot t_i (1 - p_i) \end{cases} \quad (2)$$

where, ω_i is the weight factor. The weight of the voxel closer to the surface of the vertebra is larger, while the weight of the voxel far from the vertebra is smaller. The calculation process is expressed as follows,

$$\omega_i = \gamma \cdot \exp\left(-d_i^2 / \sigma^2\right) + 1 \quad (3)$$

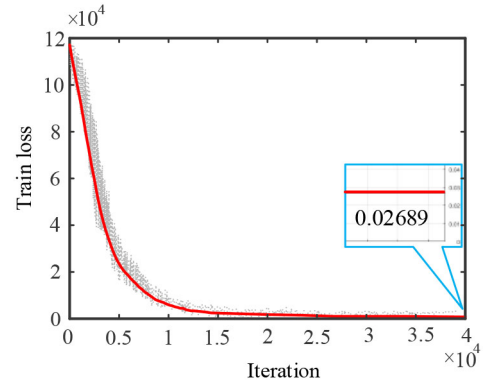


Fig. 3. Loss function changes in the network training process.

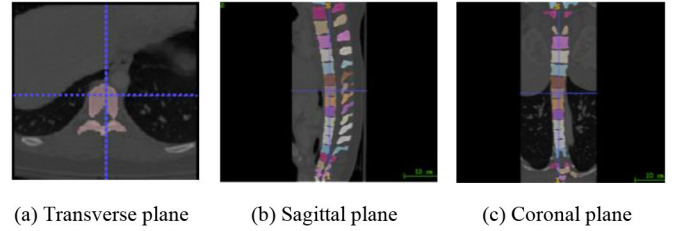


Fig. 4. Results of instance segmentation.

where, γ is 8 and σ is 6 in our experiment. In addition, for different input image data, t_i is the binary label, and p_i is the corresponding probability prediction value.

The dataset is obtained from MICCAI 2014 Computational Spine Imaging (CSI) [35], which mainly consists of several CT scans of human spines, including all thoracic and lumbar vertebrae. The scanned subjects were healthy young adults aged 20 to 34 years. During the training process, 10 sets of data were selected for training, and another 5 sets of data were selected for testing and evaluation of the segmentation results. The PyTorch framework was chosen to implement the network model, and the whole model were trained on two 8 GB NVIDIA GTX2080TI GPUs. The learning rate was set to 0.001, and the dynamic increment was set to 0.99. The training results were obtained after 4-5 days of 40,000 iterations. In the iteration process, the training parameters of the model are recorded atomically when the loss function decreases. The relationship between the loss function and the number of training iterations is drawn as shown in Fig. 3. Eventually, the loss function $\mathcal{L}$ reached 0.02689, that is, the network achieved good segmentation for the spine.

After this network model is trained, it is employed to test the segmentation of different spines. The CT scans of 10 patients that were obtained from the hospital were chosen as the test input, which included different types of spine images, such as a whole spine image, a partial spine image, and other different types. The segmentation results are shown in Fig. 4. It can be seen from the figure that, considering the high complexity and high uncertainty of the scan image, the deep learning method becomes a feasible solution to finish the segmentation task. By using a neural network, semantic segmentation is completed by identifying the type and location of the spine and predicting the exact area of the spine in the image. Furthermore, the different

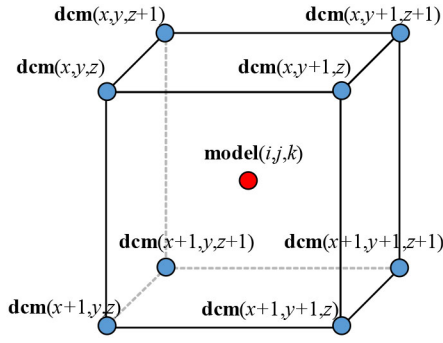


Fig. 5. Process of resampling.

segments of the vertebra are distinguished and marked differently to achieve instance segmentation. According to the labeled results, the data of each vertebra can be extracted separately, which can serve as the data basis for the next step of path planning.

B. Establishment of Local Coordinate System of Vertebra

1) *Reconstruction of 3D Model*: Before surgical path planning for the vertebra, the first thing that needs to be completed is to establish a local coordinate system to obtain a standardized description of the vertebra. In the trained results of the neural network model, labels generated by instance segmentation are used to divide different spinal segments, and three-dimensional information of the spinal segment can be extracted by using labeled volume data information. However, due to the different specifications of CT image equipment employed in data scanning, the formats of images also differ. In CT image data, there are two kinds of important parameters: layer spacing and pixel spacing. The layer spacing is the actual distance between two adjacent layers during CT tomography, while the pixel spacing is the actual distance between two pixels in the same transverse plane. To complete the 3D model reconstruction, it is necessary to resample the dimensions with large spacing according to the small spacing and to interpolate the missing point data to obtain an accurate model.

The resampling process was completed by using equations to obtain a 3D volume data model as follows,

$$\begin{cases} \mathbf{model}(i, j, k) = \sum_{p=0,1}^{p=0,1} \sum_{q=0,1}^{q=0,1} \sum_{r=0,1}^{r=0,1} A_p \cdot A_q \cdot A_r \\ \quad \cdot \mathbf{dcm}(x+p, y+q, z+r) \\ A_p = 1 - p - (1 - 2p) \cdot a \\ A_q = 1 - q - (1 - 2q) \cdot b \\ A_r = 1 - r - (1 - 2r) \cdot c \end{cases} \quad (4)$$

where, $\mathbf{model}(i, j, k)$ is the point that needs to be calculated. $\mathbf{dcm}(x+p, y+q, z+r)$ represents the 8 vertices on the nearby cube. A_p, A_q, A_r are the multipliers during the calculation. a, b are the pixel spacings in two dimensions on the transverse plane. c is the layer spacing. Through the equations, this missing point $\mathbf{model}(i, j, k)$ can be obtained. In addition, the relative of these points is shown in Fig. 5.

After resampling, the vertebral information was shown correctly and accurately. For example, when the layer spacing is larger than the pixel spacing, the obtained model will be compacted before resampling. As shown in Fig. 6, resampling

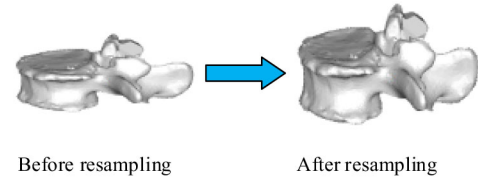


Fig. 6. Models obtained after resampling.

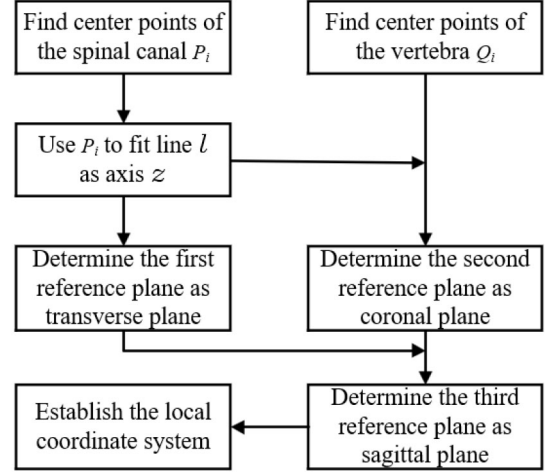


Fig. 7. Process of establishment of coordinate systems.

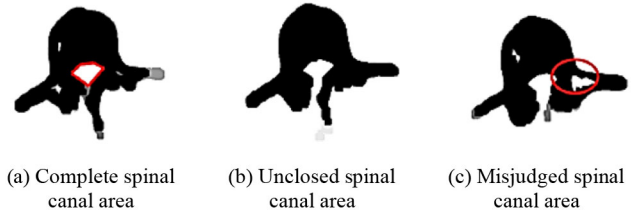


Fig. 8. Spinal canal shape at different layers.

cancels this kind of error between two different spacing parameters and obtains the correct model, which is identical to the true vertebra of the patient.

2) *Identification and Establishment of Coordinate Systems*: Based on the image information of the vertebra, the special feature points of vertebrae are recognized so that a local coordinate system can be established on the image. The steps are shown in Fig. 7.

In the transverse plane of the vertebra, the main regions that need to be recognized include the spinal canal area and vertebral body area. Commonly, the spinal canal area is always the second largest connected region. Through integration, the center points P_i of the spinal canal are determined. However, due to the different shapes of the spinal canal in continuous transverse planes of the vertebra, a non-closure situation with mistakes may occur, as shown in Fig. 8. To enhance the robustness of the algorithm, it is necessary to record the size of spinal canal areas, which can be analyzed statistically and selected to ensure that the identified center of the spinal canal is accurate.

Based on the images, data of large areas (spinal canal regions are not closed) and small areas (wrong regions) are removed among those results. The regions with highest

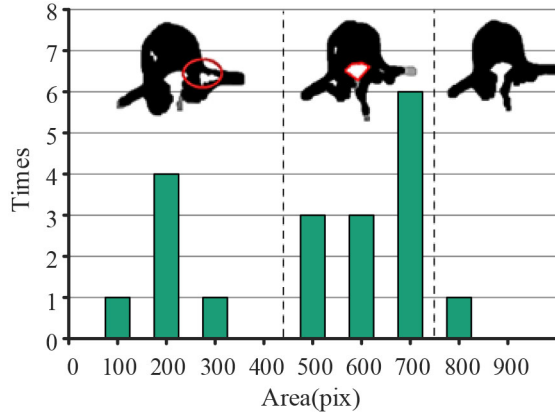


Fig. 9. Statistical result of spinal canal area.

frequency, in which the shape of the spinal canal can be seen clearly according to the frequency distribution, are chosen, making the algorithm more stable.

As shown in Fig. 9, a frequency graph is obtained after statistical analysis of the connected area of the closed interval on each transverse plane. In the section of the entire spine segment, some errors may occur near the upper and lower surfaces of the vertebral body. An accurate spinal canal region can be identified in most sections of the main body of the vertebral body, so the complete spinal canal regions are shown most frequently. By selecting the interval with the highest frequency and setting the threshold, it can be determined that different areas correspond to misjudged spinal canal areas, complete spinal canal areas and unclosed spinal canal areas. According to the complete spinal canal area section, the central point P_i was identified through the method of area integration to obtain the center of mass.

Based on the information of central points P_i , a Line l can be determined by using the least-squares method in 3D space as the Axis z , as shown in the following equations,

$$\begin{cases} x = a \cdot z + b \\ y = c \cdot z + d \end{cases} \quad (5)$$

where, a , b , c and d are the parameters of the line, which can be calculated by using the coordinate information of P_i , as shown in the following equation,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n P_{ix}P_{iz} & \sum_{i=1}^n P_{ix} \\ \sum_{i=1}^n P_{iy}P_{iz} & \sum_{i=1}^n P_{iy} \end{bmatrix} \begin{bmatrix} \sum_{i=1}^n P_{iz}^2 & \sum_{i=1}^n P_{iz} \\ \sum_{i=1}^n P_{iz} & n \end{bmatrix}^{-1} \quad (6)$$

where, P_{ix} , P_{iy} and P_{iz} are three coordinates of the spinal canal center point P_i and n is the number of spinal canal center points P_i . After obtaining Axis l , the plane perpendicular to Axis l is determined as the transverse plane of this vertebra, which is the transverse plane of the vertebra, as shown in Fig. 11.

In addition, a Voronoi diagram was utilized to calculate the maximum inner circle in each transverse plane of the vertebra to determine the center point Q_i of the vertebra. In this method, Voronoi edges divide the plane into polygon regions, each containing only one point in the point set. The intersection of three Voronoi edges forms a Voronoi point, that is, the center of the maximum inner circle of the three points forming the three

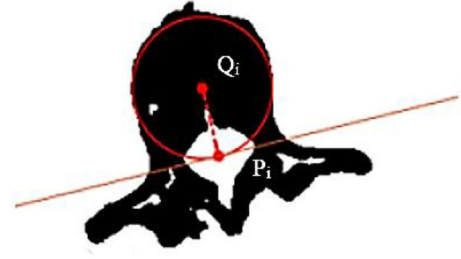


Fig. 10. Center point of spinal canal P_i and the center point of vertebra Q_i .

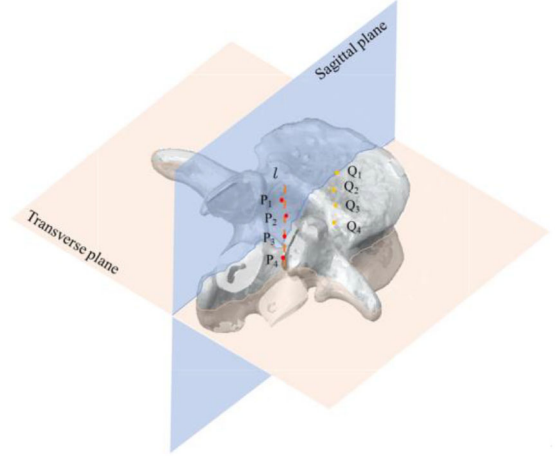


Fig. 11. Determination of coronal plane of vertebra.

sides. Using the property of Voronoi points, the maximum inner circle of the whole graph can be obtained by traversing Voronoi points. For the section of the vertebral body, the sampling points were reduced to less than one-tenth by deleting irrelevant data points, and then the Voronoi diagram was established, in which the maximum inner circle of the vertebra was obtained. The center of this circle is the center point of vertebra Q_i . The line that goes across the center point of the spinal canal P_i and the center point of the vertebra Q_i can be employed as the axis of symmetry in this image, as shown in Fig. 10.

A series of spinal canal center points and vertebra center points can be obtained through different layers of CT images. The next step is to calculate the shortest distance from the centre points Q_i to the planes that contain Axis l and to take the sum of the distances as the optimization goal to make the plane as close as possible to all the centre points of the vertebra. The equations are presented as follows,

$$x - a \cdot z - b + h \cdot (y - c \cdot z - d) = 0 \quad (7)$$

where, a , b , c and d are the parameters of Line l , and h is the parameter of those planes containing Axis l . These parameters can be obtained by the following equation,

$$\text{dis}_{\min} = \min \left(\sum \frac{|-Q_{ix} - h \cdot Q_{iy} + (a + h \cdot c) \cdot Q_{iz} + b + h \cdot d|}{\sqrt{1 + h^2 + (a + h \cdot c)^2}} \right) \quad (8)$$

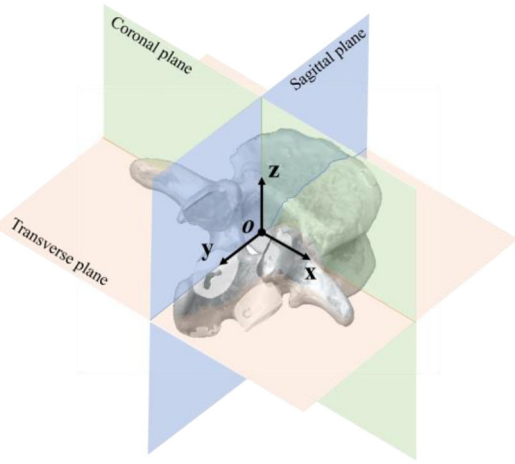


Fig. 12. Local coordinate system of a single vertebral body.

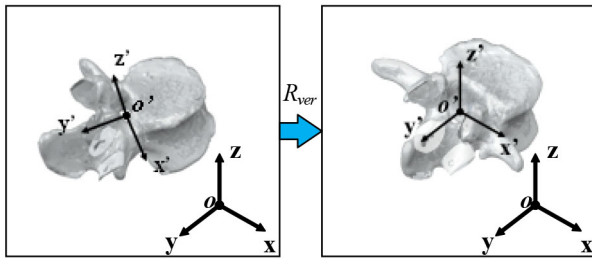


Fig. 13. Rotation of a single vertebra.

where, Q_{ix} , Q_{iy} and Q_{iz} are three coordinates of the vertebra centre points Q_i . Through the calculation, a dis_{min} which minimizes the optimization goal to obtain the plane that is closest to all points Q_i is determined. The symmetry plane of the vertebra, which is also the sagittal plane, is determined as shown in Fig. 11.

The direction and position of the third reference plane of this coordinate system can be conveniently and directly calculated according to the first and second reference planes to establish the local coordinate system. A centre point of the spinal canal that is nearest to the centre of the whole vertebra is selected as the origin of this coordinate system, as shown in Fig. 12. The vertebra is put into the coordinate system for description, which makes it convenient to identify the shape characteristics and plan for the surgery path in the following steps.

C. Planning of Screw Placement Path

In the operation of the pedicle screw, it is necessary to ensure that the screw is placed in the centre of pedicle regions, so identification of the pedicle region is the key to selecting points of the pedicle screw path. Based on the local coordinate system, the orientation of the vertebra is corrected through the rotation matrix R_{ver} , ensuring that its posture is rotated to standard directions, as shown in Fig. 13, which conforms to regular observation habits and is convenient for the calculation and planning of pedicle screw paths.

Planes parallel to the coronal plane along with the direction of $P_i Q_i$ were used at a certain distance to observe information

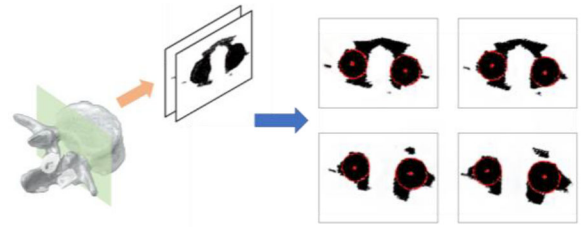


Fig. 14. Selection of pedicle region.



Fig. 15. Two parts of chosen regions.

Algorithm 1 Calculation of the Centre Points of Screw

Given: 3D model data vertebra V , coronal plane $P_{coronal}$, centre point of spinal canal C_{canal} .

For $P_i \parallel P_{coronal}$ and P_i around C_{canal}

$S_i = P_i \cap V$

For $(x, y) \in P_i$

If $x < 0.6 \cdot \text{length of } P_i$

$S_{left}^i = S_{left}^i + (x, y)$

end if

If $x > 0.4 \cdot \text{length of } P_i$

$S_{right}^i = S_{right}^i + (x, y)$

end if

find the centre point C_{left}^i of max inner circle in S_{left}^i

find the centre point C_{right}^i of max inner circle in S_{right}^i

end for

$C_{left} = \frac{\sum_{i=1}^n C_{left}^i}{n}, C_{right} = \frac{\sum_{i=1}^n C_{right}^i}{n}$

in the 3D image, as shown in Fig. 14. The shape of the vertebral pedicle region can be intuitively obtained through the observation of those planes. To locate pedicle regions on both sides, the image is divided into two parts using a regional selection algorithm. To guarantee the stability of the algorithm, some region allowance is established in both parts, as shown in Fig. 15, ensuring that pedicle regions can be included in the dividing area completely on both sides. Further calculations are carried out in the left and right parts to obtain the key points. On the cross-sections, by continuously calculating the maximum inner circle, the points closest to the centre of the vertebral pedicle regions can be obtained. The points of the two groups were collected and recorded; they were statistically processed to obtain the centre points C_{left} and C_{right} of the left vertebral pedicle region and right vertebral pedicle region. The whole process is shown as Algorithm 1.

In addition, according to anatomical knowledge of human spinal vertebrae, due to the symmetrical characteristics of the human body, the intersection point of the right and left pedicle paths is usually located on the axis of symmetry at the end of the vertebra, as shown in Fig. 16. The intersection point M

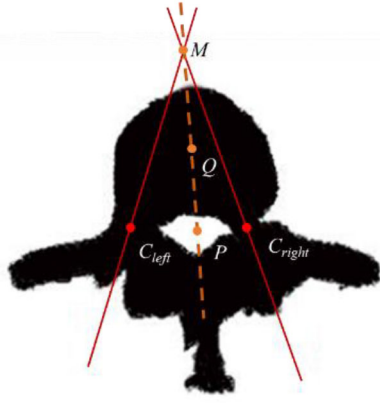


Fig. 16. Path planning of pedicle screw placement in the transverse plane.

TABLE I
DIFFERENT COEFFICIENT IN VERTEBRAE

	T10-T12	L1-L4	L5
k	1.2	1	0.7

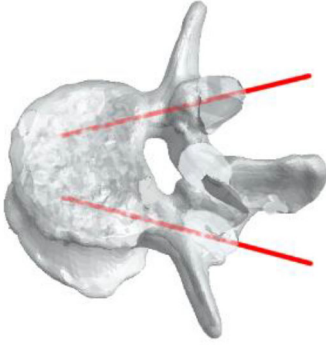


Fig. 17. Result of path planning in 3D model.

of the pedicle paths was determined according to the different sizes of vertebral bones and calculated from this equation,

$$QM = k \cdot PQ \quad (9)$$

where Q is the centre point of the vertebra, P is the spinal canal centre point, M is the intersection point of the pedicle paths, and k represents the different coefficients in different vertebrae. Due to the differences in anatomical structure characteristics of spinal segments, the value of proportional coefficient k varies in different spinal segments, and the corresponding relationship between its value and spinal segments is shown in Table I. Through the calculation, the planned points are recorded to obtain the path for screws.

The screw placement path is displayed in the 3D model as shown in Fig. 17. It can be intuitively seen from the result that the planned pedicle screw placement path is located in the central position of the vertebral pedicle region. The surgical path planning of pedicle screw placement using CT images as input data has automatically planned a good surgical approach path.

After path planning of pedicle screw placement for a single vertebra, the results need to be restored to the initial CT data. The path points recover to the initial images through

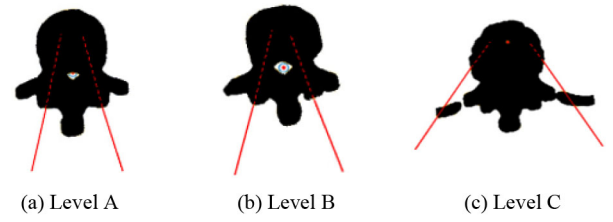


Fig. 18. Different levels of path planning results.

TABLE II
DIFFERENT LEVELS OF PATH PLANNING RESULTS

Case	A	B	C	Total
1	5	2	0	7
2	3	0	0	3
3	4	1	1	6
4	4	1	1	6
5	4	2	1	7
6	3	0	0	3
7	5	1	0	6
8	4	1	0	5
Total	32	8	3	43
Percentage	74.42%	18.60%	6.98%	100%

the rotation matrix R_{ver}^{-1} , and then segmented vertebrae are pieced together in sequence according to the serial number. Planned paths are shown in the images of the entire spine scanned preoperatively.

III. EXPERIMENTS AND RESULTS

To verify the feasibility of this method, data were obtained from Beijing Jishuitan Hospital with the consent of doctors and patients. Based on the real spine CT images of 8 patients obtained from the hospital, the proposed automatic path planning method of pedicle screw placement was carried out. Due to the different situations of different patients during CT scans, different kinds of CT images include complete or partial spines. Path planning was implemented on 43 vertebrae, and the planning results were given to professional surgeons to rank as feedback. The statistics of the planning results are shown in the following table, in which A represents excellent, B represents good, and C represents unqualified, as shown in Fig. 18.

As shown in Table II, 93% of the results show good levels, which can be well applied to the surgical path after being judged by surgeons. According to the data, the planning results are more stable at the L1-L4 lumbar vertebrae and T10-T12 thoracic vertebrae but perform poorly at the L5 lumbar vertebra or some incomplete vertebrae. The identification of feature points based on CT image data is easily affected by nearby bone areas, such as the pubis and ribs, resulting in errors in calculation. In conclusion, the path planning algorithm can quickly locate pedicle regions in most vertebrae as a highly automatic way to complete the planning of surgical paths for screw placement, which greatly simplifies the surgical process and decreases the operation time.

TABLE III
ANGLES BETWEEN AXIS OF SYMMETRY AND PLANNING SCREW PATH

Vertebral	Number	Left(°)	Right(°)	Average
T10	2	17.41±2.91	18.48±1.27	17.94
T11	5	17.71±2.17	18.68±4.35	18.19
T12	5	18.15±1.00	17.78±4.65	17.96
L1	7	14.52±3.81	15.96±3.86	15.24
L2	8	14.30±2.76	15.64±3.10	14.97
L3	6	14.99±1.72	15.60±2.10	15.30
L4	5	15.00±3.45	16.61±2.60	15.81
L5	2	15.73±2.80	15.54±2.75	15.64
Total	40	15.66	16.60	16.13

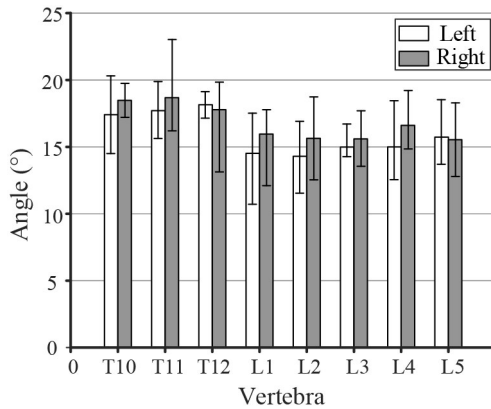


Fig. 19. Statistical result of angles between the axis of symmetry and the planning screw path.

In addition, according to the planning results of different vertebrae, a horizontal comparison is made in the same segment in different patient cases by measuring the angle between the axis of symmetry and the direction of screw insertion. Analysis of the measurement results is shown in Table III and Fig. 19.

T10, T11, and T12 are the end segments of the thoracic vertebrae, while L1, L2, L3, L4, and L5 are the lumbar vertebrae segments of the thoracic vertebrae. L1-L3 are the most frequent segments collected in CT images as diseases often occur first in the lumbar region of these patients. According to the statistical results in Table III, the angle between the axis of symmetry and the direction of screw insertion falls within the range of 10°–20°, and the entry point is located at the bottom of the facet joints. The whole path passes through the central area of the pedicle as far as possible to achieve greater holding force for screw placement, which meets the standards of screw placement path planning of surgeons [36], [37]. In the thoracic vertebrae (T10, T11, and T12), the angle was approximately 18°, while in the lumbar vertebrae, the angle was approximately 15°. The planning results are consistent with the experience of surgeons [38]. Moreover, the angles are similar in the planning of the same segment of different patients, which indicate that the proposed path planning algorithm has good adaptability and stability for different cases.

IV. DISCUSSION AND CONCLUSION

The pedicle screw technique has been widely utilized in spine surgery because of its good fixation and clinical advantage in the treatment of spinal diseases. Due to the complex shapes of the spine, which is proximal to the central nervous system, the key point to finish the surgery is choosing the entry point and determining the angle of the screw path, which usually highly depends on the physician's personal experience. Frequent intraoperative scans are needed and will introduce the risk of radiation hazards to patients and surgeons. Therefore, we proposed a method to automatically plan the required surgical path based on preoperative CT images. Through the evaluation of the planning results, the excellent and good rate reaches 93%, proving the effectiveness of the proposed planning method. In addition, due to the application of deep learning, complex CT images can be quickly and conveniently segmented to obtain the vertebra, which greatly simplifies the procedure, reduces the time spent during operations and improves the generality compared with the traditional manual method.

The path planning of pedicle screw placement is usually affected by the specificity of vertebrae. Different vertebrae always have different shapes, so it is difficult to have a universal standard method to plan for all kinds of vertebrae. In the method proposed in this paper, the vertebra can be completely segmented according to the instance segmentation method of deep learning so that a local coordinate system is established in each vertebra. After extraction and recognition of feature points, paths of pedicle screw placement are determined for each vertebra. Based on CT images of 8 patients, the angles between the axis of symmetry and the direction of screw insertion were approximately 18° in the thoracic vertebrae (T10, T11, and T12) and approximately 15° in the lumbar vertebrae. The results are consistent with the experience of surgeons and are acceptable for clinical surgical planning.

However, there are still some shortcomings in the method proposed in this paper. The planning of the screw insertion path focuses on the angle of the surgical paths, and further research on safe regions can be carried out to limit the depth of the screw insertion. In addition, more research on the holding force and fatigue strength of screw insertion in the planning results is needed.

In this paper, the path planning experiment of pedicle screw placement was completed by inputting patients' CT image scans; the results show that planned paths meet the requirements of clinical application. The proposed algorithm can quickly obtain the expected path planning results from the input data, greatly reducing the dependence on the surgeon and simplifying the operation procedure. In addition, due to the application of deep learning and image algorithms, we address the differences between two different individuals, which has good prospects for application in clinical surgery.

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